



Minimax Network Flow Model for Assessing Impacts of Airport Capacity Perturbations

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Policy decisions that target effective utilization of the national airspace often relate to the capacities of individual airports in the system, and decisions to increase or reduce capacity (e.g., via slot restrictions) must be supported quantitatively. It is of particular interest to assess global impacts of capacity adjustments, because addressing congestion issues at one airport may have a “ripple effect” on the utilization of capacity elsewhere in the network. In this paper, an analytical model was developed that addresses system-wide implications of capacity decisions at the level of interactions between Operational Evolution Partnership 35 airports. Two main elements comprise the model: 1) a network flow model that describes both the capacities of Operational Evolution Partnership 35 airports and the demand for aircraft serving origin–destination traffic across the network and 2) a novel mathematical program referred to as the minimax node utilization problem that, when solved, minimizes the worst-case capacity utilization across all of the airports in the network. Both elements combine to allow quick assessment of both local and global effects of airport capacity adjustments. The analytical model is evaluated via case studies, including two capacity reduction scenarios and one capacity expansion scenario, focusing on the model’s ability to describe how capacity influences airport utilization throughout a network representing actual origin–destination demands in the airspace. The results from these scenarios indicate that the model provides a simple and effective technique for describing impacts of airport capacity perturbations, providing valuable high-level insight into airport utilization patterns.

Nomenclature

A	=	arc set
I	=	set of origin airports
i, j	=	airport indices
K	=	commodity set
k	=	commodity index
N	=	airport set
$\mathcal{N}$	=	node-arc incidence matrix
O	=	set of destination airports
r	=	strictly constrained airport capacity
SC	=	flow-constrained airport set
u	=	airport design capacity
x	=	flow allocation

I. Introduction

REGULATORY authorities can employ a variety of mechanisms, such as congestion fees and slot restrictions, to shape the

utilization of airspace resources. To effectively implement these mechanisms, their impacts need to be understood better, both locally (in the specific region of airspace affected by the mechanism) and globally (throughout the air transportation network). This paper develops a multicommodity network flow model to analyze the local and global utilization impacts of airport capacity perturbations, toward improving the utilization of airport capacity to mitigate congestion-related delays.

Imbalances between airport capacity and demand for airport resources contribute substantially to delays in air travel. In 2010, almost 64% of delays in the National Airspace System (NAS) were caused by weather, with greater than 26% caused by volume issues [1]. These delays result in significant economic costs to passengers and airlines; in 2007, air travel delays were determined to have cost the U.S. economy approximately \$ 31.2 billion [2]. Although improvements have been observed, in 2009, 18 out of 100 air passengers experienced some type of trip disruption, averaging 105 min per passenger [3]. In addition, the projected growth of metropolitan areas continues to impact existing airports and the number of congested airports continues to expand [4].

To meet the potential rising demand on NAS airports, the Federal Aviation Administration (FAA), and other NAS stakeholders need to engage in strategies aimed at making better use of available airport capacity over the long term. At some of the busiest airports in the NAS [such as New York LaGuardia International Airport (LGA)], it is nearly impossible to simply add capacity via a new runway [4], requiring other approaches to resolve capacity-demand conflicts at these locations. Peak-period congestion pricing, which adjusts aircraft landing fees by time of day, is one suggested method to control demand at congested airports [5]. In practice, however, conflicts have arisen due to pricing out general aviation from access to airport resources and a general lack of market principles when implementing congestion pricing [6].

Another alternative, airport slot restrictions, places hard constraints on hourly operations at individual airports by controlling the

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number of slots (time intervals for an arrival/departure for a set of dates) that are allocated to airlines and other aircraft operators. Slot restrictions have been used at some of the busier airports in the NAS since the 1960s. The experience with slot restrictions at these airports has been mixed, particularly at the New York Metroplex airports—LGA, John F. Kennedy International Airport (JFK), and Newark International Airport (EWR) [7,8]. The most recent form of slot restrictions in place at the New York Metroplex airports has been observed [9] to reduce the frequency and length of delays at those airports. Historically, however, the systems in place for allocating slots at slot-restricted airports have been criticized for not promoting competitive outcomes [10] and for the negative consequences of use-it-or-lose-it provisions [11]. Market mechanisms, such as slot auctions (see, e.g., [12,13]), have been suggested as a fair and efficient alternative to the current slot allocation system which can accommodate existing levels of passenger demand while utilizing less of the available airport capacity [11].

Although the actual or potential effectiveness of these mechanisms is well-documented, their impacts have mostly been studied on a local level, that is, the effects on traffic, delays, and airline behaviors at the airport where the mechanism is implemented. However, the effects of any perturbation of airport operations extend beyond the directly affected airport. For example, studies of delay propagation (see, e.g., [14,15]) have investigated the extent to which flight delays at individual airports result in delays in flights affecting other airports. Hence, in addition to the local benefits observed as a result of mechanisms such as slot restrictions, there are effects at other airports which need to be understood.

This work focuses on analyzing the system-wide utilization of airspace resources and how their utilization is affected by air transport policy decisions that perturb the capacity of airports. It augments existing research on quantifying the extent to which these perturbations change system-wide utilization and other operational metrics, under both present and future levels of demand, as a critical step toward evaluating air traffic management policy alternatives. For example, Chatterji and Zheng [16] use simulation models to show that reducing capacity at individual airports largely increase total system delay, though the impacts are more significant depending on which airport's capacity is reduced (reductions at Atlanta Hartsfield–Jackson International result in the greatest increase in total system delay) and the impacts are greater on particular airports than others (due to network effects). These findings provide a starting point for characterizing NAS-wide impacts of all possible capacity reduction scenarios.

The utility of a minimax formulation has recently been demonstrated by Vaze and Barnhart [17] toward the investigation of complex effects in the utilization of airport resources. The focus of that work is on quantifying NAS system delays and the potential reductions achievable, relative to the existing state of the NAS. The present work shares a minimax utilization objective with the work of Vaze and Barnhart [17], although focusing on a different issue: capturing the shifts in airport capacity utilization in response to changes in one set of parameters in the system, the design capacities of selected NAS airports.

This paper formulates a model for investigating how airport capacity perturbations shift the utilization of airport resources throughout the NAS, at the level of the physical flow of aircraft and the organization of the underlying flight network. This model, the minimax node utilization problem (MNUP), determines the minimum congestion (as measured by the maximum utilization of individual airport design capacity) for a given network of airports and sets of airport capacities and aircraft origin–destination demands, over a given time period of operation. The main objective of this analysis is to examine the capabilities of the MNUP in capturing local and global utilization responses to airport capacity adjustments produced by some policy mechanism or change in operational conditions. In demonstrating the effectiveness of the MNUP in identifying trends in the local and global effects of airport capacity perturbations, this work represents the initial stage of the development of a technique that FAA decision makers might use, in conjunction with other techniques, to evaluate potential air traffic

management policy alternatives over a variety of future NAS capacity scenarios.

The analysis begins with a base case for a typical hour of operation in a network consisting of the busiest airports in the NAS, with the goal to understand how origin–destination demand is accommodated under the MNUP. With the model, airports with substantial unused capacity relative to their origin–destination demand tend to accommodate more hub operations (operations neither originating from nor destined to an airport) than airports that are heavily utilized as a result of origin–destination demand. From the state of NAS operations provided by this base case, three separate capacity perturbation scenarios are evaluated: two where capacities are decreased at airports across a given region of the NAS and one where capacities are expanded at airports across the NAS. The evaluation of these scenarios is meant to demonstrate the utility of the MNUP in characterizing the effects of changes in airport capacity and their implications for the broader utilization of NAS resources. The results from these capacity perturbation scenarios confirm the merits of the MNUP as a technique for describing how airport utilization shifts when a key property of a given airport network, the capacities of some of its airports, are adjusted.

Section II proposes the operational concept for a minimax node utilization model for describing airport capacity perturbations, detailing the assumptions of this modeling technique, and the formulation of the MNUP. The methods used in the MNUP analysis of capacity perturbations, including the selection of the airport capacity perturbation scenarios to be analyzed, are presented in Sec. III. Section IV provides a summary of the results. Finally, Sec. V concludes the paper by reviewing the results and primary contributions of this work.

II. Operational Concept for a Minimax Node Utilization Model for Airport Capacity Management

A. Operational Concept Assumptions

This work is primarily concerned with the development of a tractable model that can characterize changes in airport utilization throughout the NAS that result from changes in capacity at a small subset of airports. The MNUP is presented as a technique for describing how the system accommodates a given level of origin–destination demand for a fixed model of network connectivity as airport capacities are adjusted. As the emphasis of the analysis is on how the underlying features of the airport network itself influences utilization, the MNUP is formulated as a model of NAS operations where demand is served by an aggregate “airline” under a system utilization objective without competition between and economic decision-making by multiple airlines. The focus here is not on the specific details of the operations, but the potential utilization of the NAS for a set of actual demands and available nonstop connections.

As a consequence, a number of assumptions have been made. In particular, competitive and economic considerations are outside the scope of this paper. Although there are no distance-based costs associated with the nonstop connections, the determination of which connections exist, as described in Sec. III, mitigates this assumption by specifying that inclusion is dependent on the completion of flights within a given time window. Additionally, aircraft origin–destination demand, or the demand to transport aircraft from origin to destination, is used as a surrogate for passenger demand and is represented by actual completed flights by major domestic scheduled air carriers, as described in Sec. III.

Because demand corresponds to historical scheduled domestic flights by major reporting air carriers, the inputs to the MNUP evaluations are based only on domestic commercial flight operations. Consequently, international flights, cargo, general aviation, and other flight operations are not incorporated. Their eventual inclusion (by adding these operations to the demands on a given input network) may alter some of the MNUP solutions presented here, but this work is diagnostic, not predictive in nature; its purpose is to assess the effectiveness of MNUP as a technique for characterizing utilization responses to changes in airport capacity, so that the specific MNUP solution is not as relevant as the nature of the changes in solutions

between scenarios. For the same reason, the MNUP formulation does not restrict solutions to be integer only, and assumes the possibility of “fractional” operations. This assumption is also made with a view toward the ultimate application of the MNUP in planning for and evaluating the full range of impacts of potential air traffic management policy options over different future NAS scenarios. If the MNUP is used for such strategic aims, the exact values of its solutions are meant to correspond to average air traffic operations for the particular time period under consideration.

B. Minimax Node Utilization Problem

An airport network is given by the multicommodity flow network defined on the connected and directed graph $G = (N, A)$, with node set N corresponding to the set of airports and arc set A corresponding to the available nonstop connections between airports (aggregated across domestic scheduled carriers). Each arc in the arc set A is defined as a unique ordered pair (i, j) such that $i, j \in N$ and $i \neq j$. Each airport i in the set N is associated with a prespecified design capacity u_i , defining a nominal capacity on the number of hourly operations accommodated by the airport, based on the key attributes of the airport (e.g., runway configuration). Within the airport (node) set N , there is a subset $SC \subset N$ of slot-restricted airports such that each $i \in SC$ is associated with the restricted capacity r_i , such that $r_i \leq u_i$, which is an upper bound on hourly operations enforced by a slot restriction or similar regulatory mechanisms.

The MNUP concerns the transport of $|K|$ commodities (where $|W|$ denotes the cardinality of set W), corresponding to the transport of aircraft from an origin to a destination, over the network defined by G to minimize the maximum utilization of design capacity u_i of any airport $i \in N$. Each commodity $k = (s_k, t_k, d_k) \in K$ is associated with an ordered pair of airports (s_k, t_k) , requiring transport of an aircraft originating at s_k and ending at t_k . Associated with each commodity is the demand vector d_k whose s_k th entry is nonnegative, t_k th entry is the negative inverse of the s_k th entry, and remaining entries are zero, specifying the exact number of aircraft to be transported from s_k to t_k , which either depart or arrive within the given hour of operation. Denoting the flow of commodity $k \in K$ on arc $(i, j) \in A$ as x_{ijk} and defining $x_k = \{x_{ijk}\}_{(i,j) \in A}$, the MNUP formulation is given by

$$\min_{\{x,z\}} \quad (1)$$

subject to

$$\sum_{k \in K} \sum_{j \in I(i)} x_{jik} + \sum_{k \in K} \sum_{j \in O(i)} x_{ijk} \leq zu_i, \quad \forall i \in N \setminus SC \quad (2)$$

$$\sum_{k \in K} \sum_{j \in I(i)} x_{jik} + \sum_{k \in K} \sum_{j \in O(i)} x_{ijk} \leq r_i, \quad \forall i \in SC \quad (3)$$

$$\mathcal{N} x_k = d_k, \quad k \in K \quad (4)$$

$$x_{ijk} \geq 0, \quad \forall (i, j) \in A, \quad \forall k \in K \quad (5)$$

where, for each $i \in N$, $I(i) = \{m \in N | (m, i) \in A\}$ and $O(i) = \{m \in N | (i, m) \in A\}$, the set of potential origin airports destined for airport i and the set of potential destination airports from airport i , respectively. The auxiliary variable z denotes the worst-case design capacity utilization for all airports, and $\mathcal{N}$ is a node-arc incidence matrix [such that $\mathcal{N}_{ij} = +1$ if arc (i, j) is directed away from node (airport) i , $\mathcal{N}_{ij} = -1$ if arc (i, j) is directed toward node i and $\mathcal{N}_{ij} = 0$ otherwise]. The constraints (2) and (3) restrict the total number of operations handled by an airport for the given hour (on the left-hand side) from exceeding, respectively, either the worst-case design capacity utilization with respect to that airport's design capacity or an enforced upper-bound on the number of total operations at that airport. From the preceding formulation, the optimal solution to MNUP, (z^*, x^*) , provides, for a given network, the minimax design capacity utilization over all airports and the set of aircraft flows associated with this minimax level.

III. Methods

To apply MNUP to analyzing the global impacts of capacity perturbations on airport utilization, a NAS-like input network is constructed and instantiated with data reflecting a “typical” day of operation within the NAS. The day chosen for this analysis is 20 January 2004, a weekday in an off-peak travel period that is chosen to be compatible with the most recent FAA capacity benchmark report [18]. The hour of this day, 1200 hrs Eastern Standard Time (EST), is the focus of the analysis, so that there is sufficient traffic to and from airports in the Pacific and Mountain Time regions. In the process of constructing the input network, a set of airports must be selected, along with the connectivity between these airports. In addition, baseline capacities for each airport, as well as baseline origin–destination demands for each pair of airports, must be determined. The combination of these features make up the *base case*, which is a point of comparison for the remaining capacity perturbation scenarios to be studied, and reflects the “true” utilization of the network for the given hour of operation under the MNUP.

The scenarios to be compared with the base case involve two cases of capacity reductions across all airports within a specific region (northeast and midwest) and a case of capacity expansion across several selected airports. In both capacity reduction cases, the region chosen includes an airport or airports where capacity restrictions have been actively implemented. For all three of these cases, the features of the input network are the same as in the base case, with the only changes being the capacities at selected airports.

A. Scenarios

1. Base Case

The input network, hereafter referred to as the network, consists of 34 airports from the FAA Operational Evolution Partnership (OEP) airports [19]. These airports serve the major domestic metropolitan areas and tourist destinations and are considered to be the busiest airports in the NAS. More than 70% of domestic passengers are transported using the FAA OEP airports, many of which serve as hubs for major air carriers. For this analysis, the 34 OEP airports in the continental United States are chosen (the only OEP airport not included is Honolulu International Airport).

For each network airport, an operational capacity may be estimated, reflecting the airport's maximum level of hourly operation, dependent on the physical configuration of its runways, weather conditions, and other operational factors. This maximum level is equal to the total number of departures and arrivals an airport can handle in an hour under routine operations if the demand exists and can be exceeded (or be an extreme overestimate) under different operating conditions. In 2004, the FAA completed its last analysis of capacity benchmarks [18] for each of the FAA OEP airports, based on input from air traffic control personnel, historical traffic operations from January 2000 to July 2002, and computations from the FAA's airfield capacity model for each airport. Capacity benchmarks were estimated using each airport's most frequent runway configuration over three different weather scenarios: optimum, marginal and instrument flight rules (IFR), along with estimates of the frequency of occurrence for each scenario. In general, the capacity benchmarks provided for each airport are a range of values, accompanied by a facility-reported capacity from each airport's air traffic personnel.

In this work, the base capacities of the 34 network airports correspond to the facility-reported capacity under optimum conditions from the FAA's 2004 analysis. In agreement with the formulation of the MNUP, it is also assumed that each airport has balanced operations (i.e., arrivals and departures are handled equally), so that there is a single airport capacity instead of separate arrival and departure capacities for each airport. Some of the 34 network airports, such as Washington Dulles International (IAD), do not have balanced operations. However, the capacity benchmarks generally reflect balanced operations across all airports, which is consistent with the MNUP formulation. Table 1 lists each of the 34 network airports along with their base operating capacities.

The connectivity between the 34 network airports is determined by considering the available flights between them over the five-hour

time window of 1000 hrs to 1500 hrs EST on 20 January 2004 (where 1200 hrs EST is the third hour). These requirements prevent most east-to-west nonstop connections in the network from being included. Flight information is taken from the Airline On-Time Performance database [20] provided by the Bureau of Transport Statistics, which provides scheduled and actual departure and arrival times for nonstop domestic flights by major air carriers. If at least one flight is completed within the time window between a pair of airports in the network, there is a link between those two airports. Because this analysis is concerned with all operations between the airports, flights between one of the 34 network airports and an airport outside the network (and vice versa) are all treated identically and assumed to only utilize a “dummy” link between the specific network airport and external airport.

For this analysis, origin–destination demand between airports concerns the physical movement of aircraft and corresponds to an aircraft trip between a pair of airports, which may or may not be direct. Demand is determined by the scheduled flights reported during the 1200 hrs EST hour of 20 January 2004, from the Airline On-Time Performance database [1]. Historically, these flights are direct, nonstop flights, but in the MNUP are units of demand to transport aircraft from an origin to a destination. Under the MNUP, these units of demand may end up being served by a sequence of direct, nonstop flights instead of a single flight. These scheduled flights reflect the existence of actual passenger demand to travel between a pair of airports during a specific hour, as well as a demand to transport aircraft between those airports.

Origin–destination demand between any pair of network airports is computed by counting the flights between the pair that are reported by the Airline On-Time Performance database either originating or terminating within the 1200 hrs EST hour of January 2004. Similarly, demand between a network airport and any airport outside the network is computed by counting such flights in the Airline On-Time

Performance database, which either originate or terminate within the 1200 hrs EST hour of 20 January 2004. Summing these flights over all airports in the Airline On-Time Performance database that are not included in the network produces an “external” demand for each network airport. This demand is added to the total flow through the given network airport, as determined by the MNUP solution, and used to compute the airport’s capacity utilization.

2. Northeast Capacity Reduction Case

The Northeast Capacity Reduction Case focuses on the effects of reducing capacity at airports in the northeastern United States. The airports affected are Boston Logan International (BOS), Baltimore–Washington International (BWI), Ronald Reagan National (DCA), EWR, IAD, JFK, LGA, Philadelphia International (PHL), and Pittsburgh International (PIT). The capacity reductions enforced on these airports adjust their capacities to their facility-reported marginal and IFR capacities provided in the FAA’s 2004 capacity benchmarks. These reductions reflect actual below-optimum levels of operation at these airports; all of the northeast airports experience either marginal or IFR conditions greater than 10% of the time. The northeast airports and their marginal and IFR capacities are provided in Table 2, all remaining airports retain their base capacities from Table 1 throughout this case.

3. Midwest Capacity Reduction Case

The Midwest Capacity Reduction Case focuses on the reduction of capacity at a set of airports in the northern portion of the midwestern United States. The midwest airports affected in this case are Cleveland Hopkins International (CLE), Cincinnati–Northern Kentucky International (CVG), Detroit Metropolitan (DTW), Chicago Midway (MDW), Minneapolis–St. Paul International (MSP), Chicago O’Hare International (ORD), and Lambert St. Louis International (STL). As with the Northeast Capacity Reduction Case, the capacities at each of these airports are reduced to their facility-reported marginal and IFR capacities in separate experiments. Again, these reductions are representative of actual below-optimum levels of operation at these airports; the only midwest airports which experience either marginal or IFR conditions with less than 20% frequency are MDW and ORD. Table 3 provides the marginal and IFR capacities of all midwest airports, with all remaining airports retaining their base capacities from Table 1 throughout this case.

4. Capacity Expansion Case

Several of the network airports had undertaken plans to expand capacity at the time the 2004 capacity benchmarks were published and were at or near completion by this date. This case considers some of the airports whose expansion plans were expected to be completed by no later than 2006. The airports affected by expansion are Atlanta Hartsfield–Jackson International (ATL), CVG, Denver International (DEN), George Bush Houston Intercontinental (IAH), MSP, and STL. Although BOS, CLE, Orlando International (MCO), and Miami International (MIA) also had reported expansion plans, they retain the base capacity level in the scenario because they are less

Table 1 The 34 network airports and their base operating capacities (in total operations per hour)

Airport code	Airport name	Base capacity
ATL	Atlanta Hartsfield–Jackson Intl. Airport	188
BOS	Boston Logan Intl. Airport	123
BWI	Baltimore–Washington Intl. Airport	120
CLE	Cleveland Hopkins Intl. Airport	80
CLT	Charlotte Douglas Intl. Airport	130
CVG	Cincinnati–Northern Kentucky Intl. Airport	120
DCA	Ronald Reagan National Airport	72
DEN	Denver Intl. Airport	210
DFW	Dallas/Fort Worth Intl. Airport	270
DTW	Detroit Metropolitan Airport	184
EWR	Newark Liberty Intl. Airport	84
FLL	Fort Lauderdale–Hollywood Intl. Airport	60
IAD	Washington Dulles Intl. Airport	135
IAH	George Bush Houston Intercontinental Airport	120
JFK	New York John F. Kennedy Intl. Airport	75
LAS	Las Vegas McCarran Intl. Airport	113
LAX	Los Angeles Intl. Airport	148
LGA	New York LaGuardia Airport	78
MCO	Orlando Intl. Airport	144
MDW	Chicago Midway Airport	64
MEM	Memphis Intl. Airport	148
MIA	Miami Intl. Airport	116
MSP	Minneapolis–St. Paul Intl. Airport	120
ORD	Chicago O’Hare Intl. Airport	200
PDX	Portland Intl. Airport	120
PHL	Philadelphia Intl. Airport	104
PHX	Phoenix Sky Harbor Intl. Airport	128
PIT	Pittsburgh Intl. Airport	160
SAN	San Diego Intl. Airport	56
SEA	Seattle–Tacoma Intl. Airport	80
SFO	San Francisco Intl. Airport	110
SLC	Salt Lake City Intl. Airport	130
STL	Lambert St. Louis Intl. Airport	104
TPA	Tampa Intl. Airport	105

Table 2 Northeast airport capacities for the Northeast Capacity Reduction case (in total operations per hour), with base capacities for comparison

Airport code	Airport name	Base	Marginal	IFR
BOS	Boston Logan Intl. Airport	123	112	93
BWI	Baltimore–Washington Intl. Airport	120	80	60
DCA	Ronald Reagan National Airport	72	60	48
EWR	Newark Liberty Intl. Airport	84	80	66
IAD	Washington Dulles Intl. Airport	135	120	105
JFK	New York John F. Kennedy Intl. Airport	75	75	64
LGA	New York LaGuardia Airport	78	74	74
PHL	Philadelphia Intl. Airport	102	96	96
PIT	Pittsburgh Intl. Airport	160	150	150

Table 3 Midwest airport capacities for the Midwest Capacity Reduction case (in total operations per hour), with base capacities for comparison

Airport code	Airport name	Base	Marginal	IFR
CLE	Cleveland Hopkins Intl. Airport	80	72	64
CVG	Cincinnati–Northern Kentucky Intl. Airport	120	120	120
DTW	Detroit Metropolitan Airport	184	158	136
MDW	Chicago Midway Airport	64	64	64
MSP	Minneapolis–St. Paul Intl. Airport	120	112	112
ORD	Chicago O'Hare Intl. Airport	200	200	144
STL	Lambert St. Louis Intl. Airport	104	96	64

centrally located airports within the network. The capacity benchmarks provide a single-value projection for the capacity of these airports after expansion for each level of operation (optimum, marginal, IFR). For this case, the capacity at each expanded airport is at the postexpansion projection for optimum operating conditions, and these capacities are listed in Table 4. Again, all remaining airports retain their base capacities provided in Table 1.

B. Data Collection and Analysis

The MNUP instances corresponding to the base case and each comparative scenario are implemented and solved using the Gurobi Optimizer [21]. In each case, no strictly enforced constraints on total hourly operations are imposed at any airport [i.e., the set SC from the MNUP formulation (1–5) is empty]. The base case is evaluated using the demand and connectivity for the 1200 hrs EST hour of 20 January 2004, with the capacities at all network airports corresponding to those in Table 1. For both capacity reduction scenarios, the MNUP is evaluated using the 1200 hrs EST, 20 January 2004 data, with the capacities at either northeast or midwest airports changed according to Tables 2 or 3, respectively. Separate evaluations are performed when the northeast (midwest) airports are at marginal and IFR capacities. In the capacity expansion scenario, the MNUP is evaluated using the 1200 hrs EST, 20 January 2004 data with the capacities at expanded airports changed according to Table 4. In all three cases, the results presented will focus on 1) the changes in total operations and capacity utilization at the airports where capacity is perturbed, 2) the effects of the capacity perturbations on total network operations, and 3) the changes in utilization at remaining network airports, relative to the base case and the expectations for the results.

IV. Results

A. Base Case

The MNUP solution for the base case provides the utilization of network airports, under the MNUP objective, for the aircraft demands given by the procedure described in Sec. III.A.1 and the airport design capacities from Table 1. These results reflect the “unperturbed” operations of the network under the MNUP, with

Table 4 Capacities for capacity-expanded airports in the Capacity Expansion case (in total operations per hour), with base capacities for comparison

Airport code	Airport name	Base	Expanded
ATL	Atlanta Hartsfield–Jackson Intl. Airport	188	237
CVG	Cincinnati–Northern Kentucky Intl. Airport	120	168
DEN	Denver Intl. Airport	210	266
IAH	George Bush Houston Intercontinental Airport	120	193
MSP	Minneapolis–St. Paul Intl. Airport	120	160
STL	Lambert St. Louis Intl. Airport	104	151

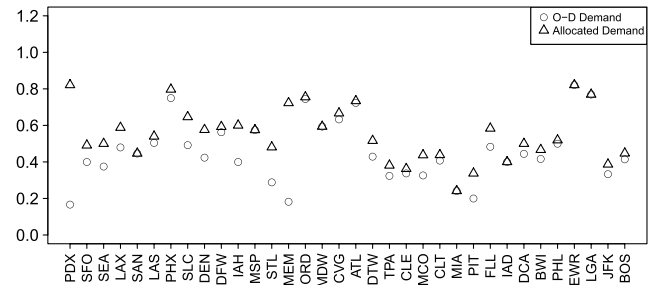
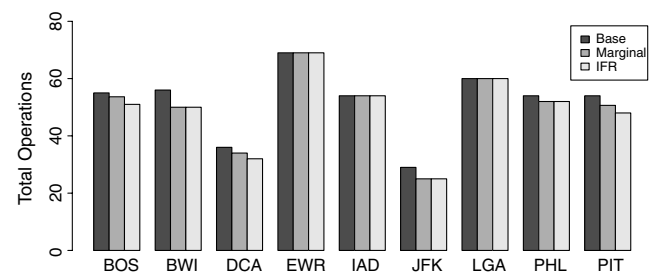


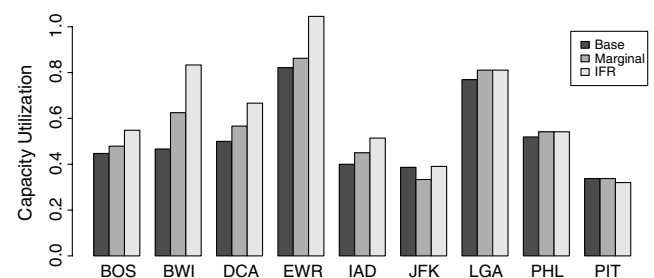
Fig. 1 Capacity utilization of all network airports in the base case.

typical demands (for the time/date in question) and best-case routine operating conditions. Figure 1 provides the base case capacity utilization for all network airports, sorted from west to east corresponding to longitude. Triangles indicate the capacity utilization allocated by the MNUP solution, circles indicate the actual origin–destination demand that each airport is required to handle, as described in Sec. III.A.1. Note that none of the network airports are overutilized and, in general, most are operating well below capacity under the MNUP solution with only domestic scheduled air carrier operations under consideration.

The base case solution also shows that the network airports whose actual origin–destination demand is high relative to their base design capacity are not allocated much additional demand by the MNUP solution. These airports include those typically considered to be “overutilized,” such as ATL, ORD, EWR, and LGA, as well as significant locations for hub operations. On the other hand, several airports with significantly higher base design capacity than actual origin–destination demand, such as IAH, MEM, PDX, and STL, accommodate substantial amounts of hub traffic. These results show that the MNUP accommodates origin–destination with an objective of providing a more balanced capacity utilization across all airports, and suggests an allocation of airport resources which is not necessarily representative of existing operations in the NAS. In doing so, however, the MNUP has demonstrated a capability to identify alternative patterns of utilization which an analyst might find interesting and worthy of deeper study. Also, the MNUP solution provides insight into how operational properties of a given network (such as the capacities of and demands on individual airports) affect network-wide utilization of resources.



a) Total operations at northeast airports



b) Capacity utilization at northeast airports

Fig. 2 Local impacts of the Northeast Capacity Reduction Case.

B. Northeast Capacity Reduction Case

The results of the base case in Fig. 1 show that several northeast airports (EWR, LGA, and IAD) only accommodate their origin–destination demand under the MNUP solution. At these airports, it should not be expected that operations should change as a result of any capacity reductions. At the remaining northeast airports, where some level of hub demand is accommodated, capacity reductions should be accompanied by reductions in operations. However, from Tables 2 and 3, the capacity reductions at some airports are not significant. For example, at PHL and PIT, the reduction from base capacity is less than 10% at the marginal level and remains there at the IFR level. Consequently, reductions in operations at airports such as PHL and PIT may be minimal and the utilization of each airport may increase. The total operations at the northeast airports should still be expected to decrease as a result of the capacity reductions in this scenario, as should the share of total network operations handled by the northeast airports.

On the network level, a reasonable expectation is that reduced operations at northeast airports are accommodated elsewhere, and the total operations in the network would at least remain the same, if not increase due to longer routes resulting from reduced capacity. Hence, an expected result is that northeast airports handle a smaller percentage of the network's total operations, after the airports' capacities are reduced. Any operations occurring at northeast airports before capacity reductions would be expected to shift to airports with some proximity to the northeast (southeast or eastern parts of the midwest) and with significant unutilized capacity (roughly 50% or greater) relative to their origin–destination demand, even though spatial aspects of air transportation have largely been ignored in the MNUP. From Fig. 1, airports such as Charlotte Douglas (CLT), CLE, CVG, MEM, MIA, MCO, and Tampa (TPA) are candidates for increased utilization at either level of capacity reduction.

Figures 2a and 2b show the changes in total operations and capacity utilization, respectively, for the northeast airports from their base capacities to their capacities at marginal and IFR levels, under the MNUP solutions. As expected, the airports which only served their origin–destination demands (EWR, LGA, and IAD) experience no changes in overall operations while their utilization levels increase with reductions in capacity. The remaining northeast airports each experience decreases in operations as a result of at least one level of reduced capacity, accompanied by general increases in utilization levels. These reductions in operations also occur at airports such as JFK, PHL, and PIT, even when their own individual airport capacities do not change. These results show that the MNUP can reveal downstream utilization responses to airport capacity changes. In this case, JFK, PHL, and PIT experience reductions in operations that may not be a direct result of their own capacity levels, but are the result of their interactions with other northeast airports and changes in operations elsewhere in the network.

Table 5 and Fig. 3 display the impacts of the capacity reductions on the network as a whole. Table 5 compares the total network operations and operations handled by the northeast airports at the base, marginal, and IFR capacities from the MNUP solutions for each level. As expected, the share of total network operations handled by the northeast airports decreases, accompanied by a decrease in total operations at these airports. A decrease in total network operations is also observed, a result of some of the utilization impacts on non-northeast network airports shown in Fig. 3. These airports are the non-northeast airports in the network that experience the most significant changes in operations from the

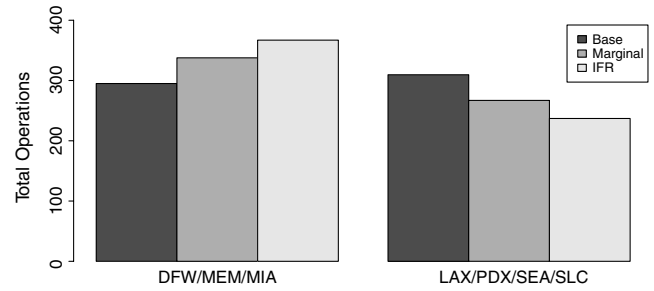


Fig. 3 Global impacts of the Northeast Capacity Reduction Case.

base case. When northeast airports operate at their marginal and IFR capacities, at least a 10% change in the total number of operations over the base case is observed at each of the airports included in Fig. 3. Dallas/Fort Worth (DFW), MEM, MIA experience the most significant increases in total operations, whereas the most significant decreases in operations occur within a group of western airports: Los Angeles (LAX), PDX, Seattle (SEA), and Salt Lake City (SLC).

MEM and MIA were both expected to be potential locations for increased operations, and DFW is another such reasonable location, being the network airport with the highest overall capacity and not located in the west. This rearrangement of aircraft flow modifies operations elsewhere, and the observed reduction in operations at western airports reflect a secondary effect of the northeast capacity reductions. The western airports lose operations in part due to shifts in utilization to the “center” of the network and the minimax objective of the MNUP ensuring that the additional volume in the center is below some threshold. As a result, the western airports lose connections to the airports most directly affected by the northeast capacity reductions, in addition to losing some operations from connections to northeast airports. The combined decrease in operations at western and northeast airports contributes most significantly to the decrease in total network operations resulting from the northeast capacity reductions. The decrease in operations at western airports and throughout the network both demonstrate the capability of the MNUP in revealing how the impacts of airport capacity adjustments propagate throughout a network to accommodate a given set of origin–destination demands.

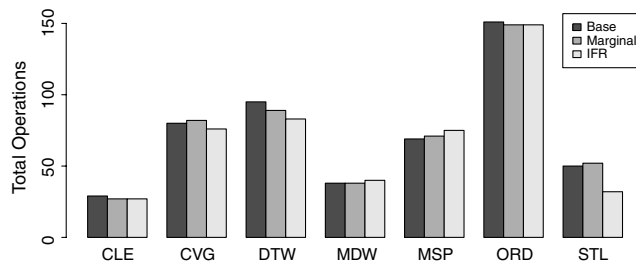
C. Midwest Capacity Reduction Case

Among the seven midwest airports (STL, ORD, MDW, CVG, CLE, DTW, and MSP), only DTW and STL accommodate significant hub traffic in the base case (Fig. 1). Note that zero hub traffic is accommodated by midwest airports MDW and MSP. From Table 3, neither CVG nor MDW lose any capacity from base to marginal to IFR levels, and Chicago O'Hare maintains its base capacity at marginal operating conditions before a steep reduction at IFR conditions. Only at DTW are the capacity reductions significant across the operating conditions, and the expected result is that the reductions in operations will be greatest here. Although the minimal reductions in capacity elsewhere may complicate the expectations for the results, decreases in the operations at the midwest airports should still generally be expected, as only two airports do not experience a capacity reduction at all and nearly all accommodate some hub traffic in the base case. However, unlike the northeast case, the presence of airports where capacity is not actually reduced allows for increases in operations at airports such as CVG, MDW, and ORD (at the marginal level). Capacity utilization at each individual midwest airport, for all the aforementioned reasons, can generally be expected to increase or remain the same.

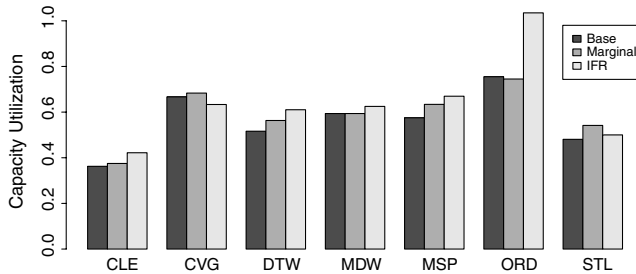
Because some of the midwest airports may actually increase operations, the total operations these airports accommodate should decrease by less than that of the northeast airports. The share of network operations accommodated by the midwest airports should also decrease, under the assumption that the reduced operations at these airports are accommodated elsewhere, either maintaining or increasing the number of network operations. Airports where

Table 5 Impacts of northeast airport capacity reductions on overall and northeast operations.

Capacity level	Network operations	Northeast operations	Northeast share
Base	2351.6	467.0	0.1986
Marginal	2290.9	448.3	0.1957
IFR	2269.0	441.0	0.1944



a) Total operations at midwest airports



b) Capacity utilization at midwest airports

Fig. 4 Local impacts of the Midwest Capacity Reduction Case.

operations may be shifted include any airport which can be considered “midwestern,” or at least longitudinally similar from the x axis of Fig. 1, and has significant (roughly 50% or greater) unused capacity relative to its origin–destination demand. Airports such as CLT, DFW, IAH, MEM, MIA, MCO, PIT, and TPA fit these criteria.

Figures 4a and 4b show the changes in total operations and capacity utilization at the midwest airports, as a result of the capacity reductions at these airports. In general, the expected increases in capacity utilization are observed. As expected, the largest decrease in operations occurs at DTW, and it is also the only midwest airport where operations decrease at each of the progressive levels of capacity reduction. The only other airports in this group which do not experience an increase in operations at either marginal or IFR levels are CLE and ORD. Both of these airports accommodated minimal hub operations in the base case, and so this result is consistent with expectations, even though the capacity at CLE declines steadily as a result of the reductions, and the capacity at ORD drops steeply from marginal to IFR levels. At the remaining midwest airports, there are small fluctuations (less than 10% of the base case operations) in operations from base levels when there is no actual reduction in capacity or the reduction is small (as in the cases of CVG, MDW, and MSP at all capacity levels, and in the case of STL at the marginal level). With no or minimal reductions, the operations at these airports could not have declined substantially, and increases reflect the accommodation of operations lost at other midwest airports experiencing larger capacity reductions. When capacity at STL is reduced more significantly to its IFR level, operations also drop by more than a third of base case operations, which coincides with STL’s accommodation of hub operations in the base case.

Because of the mix of responses to capacity reduction, the total operations at midwest airports only decreases slightly, as shown in Table 6. The share of network operations at the midwest airports, however, decreases by more than in the northeast airport case, as a result of increased network operations. The very small number of

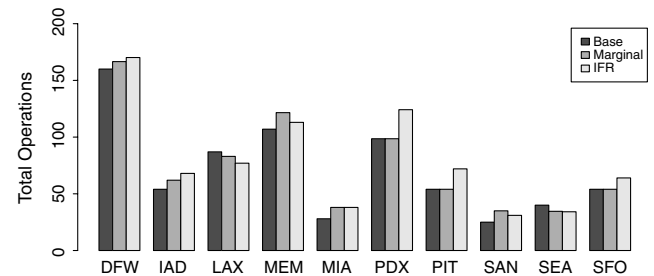


Fig. 5 Global impacts of the Midwest Capacity Reduction Case.

operations “lost” by midwest airports means that the overall increase in network operations is more of a propagated effect of the capacity reductions at those airports. Several of the midwest airports are responsible for accommodating significant east-to-west traffic, and the loss of capacity at these airports results in some longer commodity paths, which add operations at other airports.

In Fig. 5, responses to midwest capacity reductions at other airports are presented. Some of the locations expected to see increased utilization (ATL, DFW, MEM, MIA, and PIT) do increase operations but to differing extents. Several airports on both coasts [BWI, IAD, PDX, San Diego International Airport (SAN), and San Francisco International Airport (SFO)] see increases in operations as well. These increases indicate direct impacts from accommodating origin–destination demand that the midwest airports no longer serve, as well as secondary impacts that result from an increased frequency of longer commodity paths. As in the northeast capacity reduction case, operations decrease at LAX and SEA in response to the capacity reductions at midwest airports. Several of the western airports that experienced drops in operations in the northeast case, however, increase operations as a result of the capacity reductions here. In this case, the changes in utilization at these western airports reflect different effects. The airports that increase operations accommodate some origin–destination demand that has shifted, in addition to utilizing connections with other airports which have increased operations but whose utilizations do not get close to the MNUP optimal value. On the other hand, LAX and SEA experience reduced operations due to base case relationships with the midwest airports (where operations decrease) or airports such as DFW and MEM (whose operations increase, but accommodate different demand resulting from capacity reductions at midwest airports).

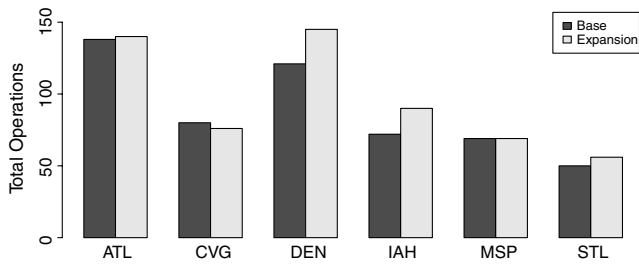
D. Capacity Expansion Case

From the base case, several of the capacity-expanding airports (DEN, IAH, and STL) already serve substantial hub operations and should be expected to serve more with additional capacity in this case. The remaining capacity-expanding airports (ATL, CVG, and MSP) accommodate little in the way of hub operations and can be expected to accommodate slightly more operations beyond the origin–destination demand each airport is required to serve. The increase in operations resulting from capacity expansion should be expected to be greater at the airports that accommodate more hub operations in the base case. The MNUP objective defines this role for these airports, whose origin–destination demands are smaller relative to their base capacities, in comparison with the airports in this group serving minimal hub operations. In general, the operations at these airports should increase, although the scale of capacity expansion (each airport in this group expands by greater than 30% over its base capacity) most likely means the increased operations may not effect an increase in capacity utilization as well.

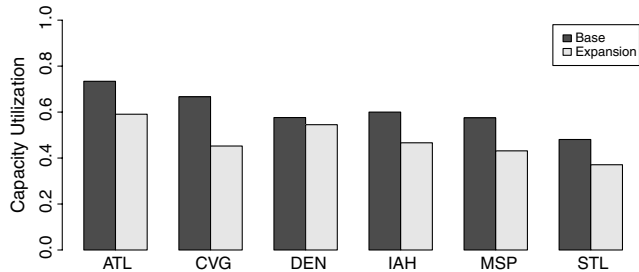
The increase in overall operations amongst the capacity-expanded airports should lead to an increase in the share of network operations accommodated by these airports because the additional operations should naturally be balanced out by decreased operations elsewhere. As a result, overall network operations should be expected to stay the same, if not decrease, because the increased capacity could potentially shorten routes for origin–destination demand. The remaining network airports most likely to be affected by capacity

Table 6 Impacts of midwest airport capacity reductions on overall and midwest operations

Capacity level	Network operations	Midwest operations	Midwest share
Base	2351.6	462.0	0.1965
Marginal	2387.3	456.0	0.1910
IFR	2415.5	450.0	0.1863



a) Total operations at capacity-expanded airports



b) Capacity utilization at capacity-expanded airports

Fig. 6 Local impacts of the Capacity Expansion Case.

expansion are those that have any quantity of hub operations in the base case, because the expanding airports are not confined within a specific region. In particular, airports whose base case utilization is mostly tied into hub operations, such as MEM, PDX, and PIT, may be expected to experience the greatest decreases in total operations.

Figures 6a and 6b present the changes in total operations and capacity utilization, respectively, on the airports where capacity expansion occurs. In Fig. 6a, the total operations expectedly increase the most at DEN and IAH, the two airports that accounted for the most hub operations in the base case within this group. Out of these airports, only CVG and MSP do not increase operations as expected, with total operations at CVG actually decreasing. In the base case, both of these airports primarily served their own origin–destination demands, with nearly 40% unutilized ed capacity at each airport, indicating that expanded capacity does not address any opportunity to accommodate additional origin–destination demand at these locations. The decline in operations observed at CVG, though small (less than 5% of its base total operations), is indicative of another effect of the capacity reductions: Although CVG is in the group of expanded airports, the expansion at other airports, which accommodate more than minimal hub operations, end up serving this demand in place of CVG.

From Table 7, the overall increase in operations at the capacity-expanded airports is accompanied by an increase in their share of the total network operations. Much of the contribution to the increase in this share comes from a decline in total network operations, suggesting that the capacity added to this group of airports both decreases the load on airports outside this group and provides more available alternatives for routing origin–destination demand, which are actually utilized. To that point, Fig. 7 demonstrates that, regardless of region, network airports accommodating hub operations in the base case generally lose operations in the face of capacity expansion at the six airports under consideration. At MEM, however, a greater than 10% increase in operations is observed, which indicates a downstream effect of the increase in operations of at least one of the capacity-expanded airports (IAH, in particular).

Table 7 Impacts of capacity expansion on operations in the overall network and at capacity-expanded airports

Capacity level	Network operations	Expanded operations	Expanded share
Base	2351.6	530.0	0.2254
Expansion	2203.6	576.0	0.2614

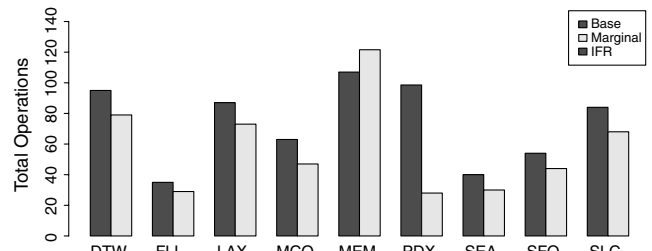


Fig. 7 Global impacts of the Capacity Expansion Case.

E. Summary of Results

The results of each of these cases, in comparison to the base case, illustrate that the MNUP analysis of airport capacity perturbations produces the expected local trends for capacity utilization changes at the airports where the perturbations occur. In response to capacity reductions, the northeast and midwest airports generally saw a decline in total operations which is not sufficient to avoid an increase in individual capacity utilizations, but is sufficient to decrease the share of total network operations handled. Conversely, in the capacity expansion case, the airports under consideration generally experienced increases in total operations, as well as their share of total network operations, however, the operational increases were small relative to the expanded capacity, leading to a decreases in the airports' capacity utilizations.

Globally, the results produced by the MNUP also effectively described the expected impacts of the given capacity perturbations. In both capacity reduction cases, increases in utilization were observed at airports where such operational changes could be classified as reasonable: in the northeast case, airports in the southeast or midwest; in the midwest case, other airports in the midwest, as well as a mix of eastern and western airports that might be required to facilitate east-to-west air traffic as a result of reduced capacity at midwest hubs. Many of the airports where utilization increases were observed also had significant unutilized capacity with respect to origin–destination demand or their base case utilizations, as might be expected from the objective of the MNUP. In the capacity expansion case, decreases in utilization were observed at many airports that accommodated some degree of hub operations in the base case, an expected response to significant expansion in capacity at airports across regions in the network.

The MNUP has also proven useful in illustrating how airport capacity perturbations propagate and produce downstream shifts in airport capacity utilization. In all cases, some network airports experienced shifts in total operations as a result of more direct impacts of the given capacity perturbation. These downstream impacts occurred both locally and globally. In the Northeast Capacity Reduction Case, some northeast airports lose operations even when their own capacities are not reduced, due to declines in operations at airports where capacity is reduced. Some western airports in this case, and in the Midwest Capacity Reduction Case, lose operations as a result of increased operations at airports that accommodate demand lost at either the northeast or midwest group of airports. In the Capacity Expansion Case, operations decrease at one capacity-expanded airport (CVG) primarily because other airports that expand capacity provide greater opportunity to accommodate hub operations. In this same case, another airport (MEM) increases operations primarily as a result of increased operations at capacity-expanded airports.

Many of these results, in addition to the results for the base case, also show that the MNUP provides insight into utilization patterns the network is capable of supporting in accommodating demand, which might be obscured in a competitive environment. In achieving the MNUP's objective, operations will shift to airports with lower origin–destination demands relative to their capacities, but in a manner that spreads operations across a group of airports and produces secondary changes in operations at other locations to prevent increases in the minimax system utilization. This type of impact is exhibited in the decreases in operations at western airports produced in the two capacity reduction cases, and the decrease at

CVG in the Capacity Expansion Case. Throughout the scenarios investigated, the MNUP has produced the expected impacts of the given capacity perturbations and revealed how these shifts in utilization produce further effects on operations throughout a given airport network. Overall, this analysis has demonstrated that the MNUP is an effective technique for describing local and global responses to airport capacity perturbations, reflecting how a given airport network accommodates demand as its properties are adjusted.

V. Conclusions

This work addresses, in part, shortcomings in the long-term management of airspace resources, specifically at the major airports in the National Airspace System (NAS). Much of the analysis for managing the utilization of busy NAS airports has focused entirely on the local impacts of policy decisions and other actions affecting air transport operations. There is a need to evaluate the impacts of these types of perturbations more comprehensively and consider their effects throughout the NAS and on the behaviors of various NAS users. Because of the complexity of the NAS, the development of techniques and tools for analyzing these system-wide impacts proceeds in increments. Different attributes of the NAS are modeled at progressive levels of fidelity to describe the full range of the potential impacts of specific system perturbations.

Toward characterizing the lower range of those impacts, a model is introduced to first describe how airport capacity is utilized across a network when the worst-case capacity is minimized and then how these utilizations shift as airport capacities are adjusted under a specified perturbation. This model, the minimax node utilization problem (MNUP), is applied to analyze operational changes for a typical hour in a network composed of the most utilized airports in the NAS and then assessed for its capability in describing these changes. Using three cases where the capacity of a subset of network airports is either reduced or expanded, a proof-of-concept is provided for the MNUP as a technique for simple evaluations of the impacts of capacity perturbations over a range of scenarios.

The MNUP provides Federal Aviation Administration (FAA) decision makers with a technique, along with other approaches of varying levels of fidelity, to quantify the effects of candidate air traffic management policy mechanisms over a range of future NAS states, with the goal to improve the selection and implementation of these mechanisms. MNUP solutions can also reveal patterns of airport capacity utilization that are achievable for given connectivity and demands, which, although not feasible in the existing system, might help an analyst in formulating mechanisms to modify the utilization of airport resources by NAS users. To accurately support these kinds of decisions, further development of the MNUP will be necessary to represent existing NAS operations and produce good estimates of the values of these capacity utilization impacts. However, at the moment, the utility of the MNUP has been sufficiently demonstrated as a technique providing simple characterizations of the impacts of airport capacity utilization and the downstream implications of these impacts in a given airport network.

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